# Literature Review: EdTech & Learning Outcomes in Africa

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## Introduction

Educational Technology (EdTech) has increasingly become an important component of modern education, particularly as digital tools and platforms worldwide provide new ways of accessing learning resources. In Africa, the growth of digital learning has created opportunities to address some educational challenges while also introducing questions about its accessibility, effectiveness and learning outcomes.
This literature review examines research on the adoption and effectiveness of EdTech in Africa, with particular attention to its influence on students' learning experiences and outcomes.

## PAPER 1 – John Edem Avedzi, 2025

*Educational Technology, Teaching Practices and Learning Outcomes in Sub-Saharan Africa: Empirical Insights from Ghana's Education Classrooms.*

Mr. Avedzi examined the influence of EdTech on teaching practices, learner engagement and academic performance of students in Ghanaian basic schools. He adopted a survey on six public schools across the Volta, Eastern, and Greater Accra Regions of Ghana. Data was collected from 300 teachers across the six public schools using questionnaires. The data were analyzed using methods such as Analysis of Variance (ANOVA) and linear regression. The study majored in learner engagement in technology use, incorporated EdTech in teaching practices and some hindrances faced.
The findings found a strong positive relationship between technology use and learner engagement, indicating learners with greater access to educational technology demonstrated better academic results. This also showed a disparity between learners from the urban and rural communities, with limited or no access to educational technology. However, poor teacher training in EdTech resulted in under utilization of available technologies. Findings also highlighted an 18% of public schools in the whole country with available I.C.T teachers, out of which 30% had access to the internet. Digital pedagogy alone, therefore did not significantly influence teaching practices, suggesting that training without continuous support and practical training may not be enough to transform EdTech use in classrooms. Overall, Avedzi concluded

that EdTech can contribute significantly to learner engagement and academic performance among students once it is properly integrated.

## PAPER 2 – Wesonga Justus Nyongesa & Johan Van der Westhuizen, 2025

*The Impact of Digital Teaching Tools on Student Engagement and Learning Outcomes in Higher Education in Africa.*

This paper examined the impact of digital tools in everyday learning. The authors reviewed existing research tools on digital tools such as Learning Management Systems, mobile applications, collaborative platforms and the like. Their findings showed that digital tools can improve student engagement, knowledge retention, problem solving and academic performance, particularly when interactive teaching strategies are used. However, challenges such as poor internet connectivity (due to server limitations, unreliable electricity), low digital literacy, institutional and mobile app challenges can reduce their effectiveness.
The study concluded that digital teaching tools have significant potential to improve EdTech in Africa higher education, but their success depends on appropriate teaching practices and adequate technological support.

## PAPER 3 – Douglas Darko Agyei, 2020

*The Impact of Educational Technology Initiatives on Student Learning Outcomes: Perspectives of Sub-Saharan Africa.*

Mr. Agyei investigated the impact of the Africa Digital School ICT initiative on student learning outcome across six sub-Saharan African countries: Ghana, Kenya, Nigeria, Tanzania, Uganda and Ethiopia. It involved 542 from the six countries, using online surveys, open-ended questionnaires and focused grouped discussions to evaluate the initiative. Findings showed that the initiative was well-received by students and improved their learning attitudes toward digital tools, but its long-term sustainability was threatened by factors such as inadequate infrastructure, limited equipment, poor internet connectivity, among others.
It was therefore concluded that educational technology initiatives required continuous support and suitable conditions for sustainable implementation and technology use.

**Challenges Affecting the Sustainability of EdTech Implementation**

| Challenge | Percentage (%) |
|---|---|
| Inadequate computers | 32.1% |
| Computer addiction/ distraction | 17.8% |
| Power outages | 13.4% |
| Access to inappropriate online content | 11.9% |
| Internet connectivity problems | 10.7% |
| Uncomfortable computer rooms | 8.6% |

***Source: Agyei (2020)***

## PAPER 4 – Innocent C. Nnorom, Solomon C. Onuchukwu, Roseline I. Ndimele & Joan I. Oji, 2026

*EdTech for Sustainable Development in Africa: Trends, Challenges, and Prospects*

The authors conducted a systemic thematic review published between 2006 and 2025 to identify major trends, challenges, and prospects of EdTech in Africa. They identified the growing use of mobile learning, artificial intelligence and adaptive learning platforms to be major trends in Africa.

The review found that EdTech has the potential to improve learning outcomes, teacher effectiveness, skills development and workforce readiness.

However, it was found that its benefits remain uneven because of major challenges inadequate infrastructure, financial constraints, limited digital literacy and some concerns about cultural and linguistic relevance. The authors concluded that sustainable EdTech integration requires investment in digital infrastructure, locally relevant content and equity challenges, as well as coordinated policy and investments.

## PAPER 5 – Abdul Badi Sayibu, Jesus Miguel Falcon Perez & Miniratu Issifu, 2026

*Improving the Foundational Literacy Skills of Marginalized Learners through Satellite-Based EdTech: The MGCubed Project in Ghana*

Sayibu, Falcon Perez and Issifu (2026) examined the impact of satellite-based educational technology on the literacy skills of marginalized learners through the Making Girls Great! (MGCubed) project in Ghana. The project provided distance learning to primarily girls aged 9-15 years from under-developed communities. Using an experimental design and Difference-in-Difference (DiD) analysis, the study compared literacy outcomes between treatment and control groups. The DiD analysis showed that learners in the treatment group scored 1.3 percentage points higher in literacy than those in the control group, a statistically significant difference at 5% level. Though not a huge jump, the observed improvement was unlikely to have occurred by chance, suggesting the project really did help improve those literacy scores.

The study concluded that satellite-based EdTech can improve foundational literacy among undeserved learners, particularly in communities where reliable internet access is limited.

## Conclusion

Overall, the reviewed studies show that EdTech has significant potential to improve learning outcomes and teaching practices in Africa. The studies found positive effects on learner engagement, foundational literacy, academic performance and teacher effectiveness. However, the benefits of EdTech are influenced by factors such as access to digital devices, internet connectivity, infrastructure, digital literacy, teacher training and supportive policies. Highlighted also were inequalities between well-served and underserved communities.

Therefore, for EdTech to achieve sustainable impact in Africa, technological adoption should be accompanied by adequate infrastructure, teacher professional development and policies that help promote equitable access to digital learning.